

NeuroSky

MindWave Mobile · Android SDK

Developer Guide

Version 2.0.1

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Real-time EEG integration for Android applications
via Bluetooth Low Energy
Built on Kotlin Coroutines & Flow

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1. Overview

The NeuroSky MindWave Mobile Android SDK is a modern Kotlin library that lets you read real-time EEG (electroencephalography) data from a NeuroSky MindWave Mobile headset on Android 6.0 (API 23) or later.

The SDK connects directly to the MindWave Mobile 2 headset over BLE (primary) or BT Classic (RFCOMM/SPP fallback), eliminating any external software dependency. Once connected, it decodes the ThinkGear ASIC data stream and exposes a `Flow<BrainWaveData>` that emits parsed EEG packets in real time.

Each `BrainWaveData` packet carries the full set of MindWave measurements: eSense Attention and Meditation indices (0-100), eight FFT frequency band powers (Delta through Mid-Gamma), raw EEG samples at 512 Hz, eye-blink intensity, and a signal quality indicator. Real-device testing with a physical MindWave Mobile 2 is required to validate BLE and BT Classic connectivity. The built-in Simulator is available as a supplementary tool for UI and logic testing only.

Key Features

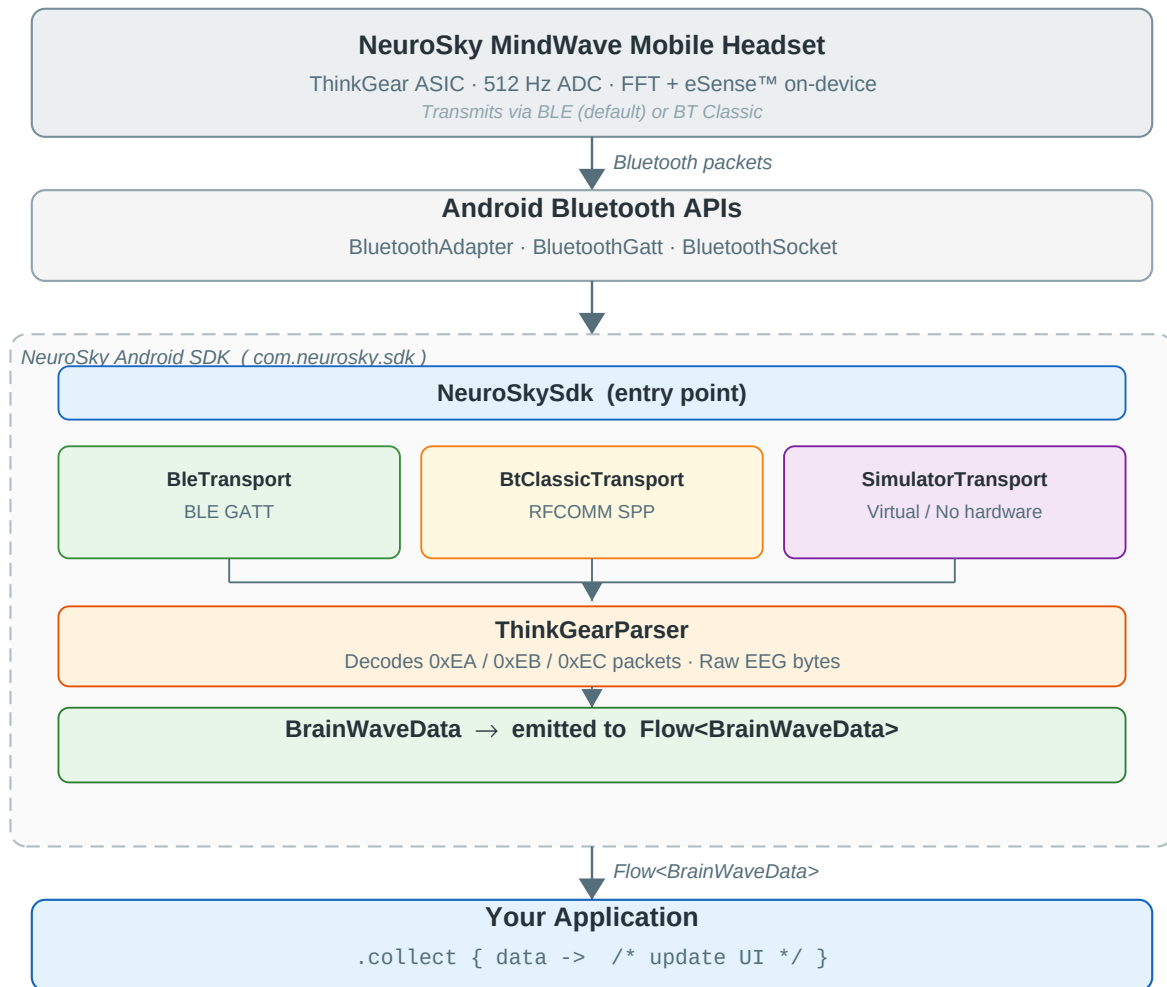
Feature	Description
BLE + BT Classic	BLE (primary) and BT Classic (fallback); real device required for full validation
Kotlin Coroutines & Flow	<code>Flow<BrainWaveData></code> · integrates naturally with Jetpack lifecycle
Built-in Simulator	Supplementary tool for UI/logic testing; not a substitute for real-device testing
JitPack distribution	One-line Gradle dependency, no local setup
Android 6.0+ (API 23+)	Wide device coverage

What You Can Measure

The MindWave Mobile headset contains a single dry electrode on the forehead (FP1) and a reference clip on the ear. The ThinkGear ASIC chip on board computes:

- Raw EEG waveform — 512 samples/sec, signed 16-bit values
- 8 frequency band powers — Delta, Theta, Alpha (Low/High), Beta (Low/High), Gamma (Low/Mid)
- eSense™ Attention — NeuroSky's proprietary attention index (0~100)
- eSense™ Meditation — NeuroSky's proprietary relaxation index (0~100)
- Eye blink detection — intensity 0~255
- Signal quality — 0 (perfect contact) to 200 (no signal)

2. How It Works — Architecture



BLE path (default): Subscribes to GATT characteristics. No device pairing required.

BT Classic path: Opens RFCOMM BluetoothSocket. Device must be paired in Android Settings first.

Both paths: Identical BrainWaveData output. No automatic fallback between transports.

3. Requirements

Device Requirements

Component	Minimum
Android OS	Android 6.0 (API level 23)
Kotlin	1.9+
Coroutines	kotlinx-coroutines-android 1.7+
Bluetooth	BLE adapter (for BLE mode) or Classic BT (for BT Classic mode)
Pairing	Not required for BLE; required for BT Classic

Supported Headset

This SDK is designed and tested for the NeuroSky MindWave Mobile 2. Both BLE and BT Classic modes are supported.

4. Installation

Step 1 — Add JitPack to your repository list

In settings.gradle.kts :

```
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
        google()
        mavenCentral()
        maven { url = uri("https://jitpack.io") } // add this
    }
}
```

Step 2 — Add the dependency

In app/build.gradle.kts :

```
dependencies {
    implementation("com.github.nsk-bci:mindwave-sdk-android:v2.0.1")
}
```

Step 3 — Sync and verify

Android Studio → File → Sync Project with Gradle Files, or via terminal:

```
./gradlew assembleDebug
```

If the build succeeds without errors, the SDK is correctly installed. Note: JitPack builds the library on first request. The very first ./gradlew run may take 30–60 seconds while JitPack fetches and compiles the source. Subsequent builds use the cached artifact.

5. Permissions

Android requires Bluetooth permissions in AndroidManifest.xml. The required permissions differ by Android version.

AndroidManifest.xml

```
<!-- Android 12+ (API 31+) -->
<uses-permission android:name="android.permission.BLUETOOTH_SCAN"
    android:usesPermissionFlags="neverForLocation" />
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />

<!-- Android 6.0-11 (API 23-30) -->
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
```

Adding all of the above covers all Android versions from 6.0 to 14+. The OS ignores permissions not applicable to the current version.

Runtime permission request (Android 12+)

On Android 12 and above, BLUETOOTH_SCAN and BLUETOOTH_CONNECT are runtime permissions — the user must grant them at runtime, not just at install time.

```
// In your Activity or Fragment
private val bluetoothPermissionLauncher = registerForActivityResult(
    ActivityResultContracts.RequestMultiplePermissions()
) { permissions ->
    val allGranted = permissions.values.all { it }
    if (allGranted) { startEegSession() }
    else { showPermissionDeniedMessage() }
}

private fun requestBluetoothPermissions() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
        bluetoothPermissionLauncher.launch(arrayOf(
            Manifest.permission.BLUETOOTH_SCAN,
            Manifest.permission.BLUETOOTH_CONNECT
        ))
    } else {
        // Android 11 and below: request location permission
        bluetoothPermissionLauncher.launch(
            arrayOf(Manifest.permission.ACCESS_FINE_LOCATION)
        )
    }
}
```

Why location permission is needed (Android 6-11)

On Android versions before 12, BLE scanning is considered a location-revealing operation because BLE beacons can be used for indoor positioning. The OS therefore requires ACCESS_FINE_LOCATION to perform BLE scans — even if your app has nothing to do with location. This requirement was lifted in Android 12 with the introduction of BLUETOOTH_SCAN.

6. Quick Start

The following example is a complete minimal implementation inside an Activity using lifecycleScope :

```
import com.neurosky.sdk.NeuroSkySdk
import com.neurosky.sdk.NeuroSkyCommand
import com.neurosky.sdk.model.SignalQuality
import com.neurosky.sdk.transport.ConnectionState

class MainActivity : AppCompatActivity() {
    private lateinit var sdk: NeuroSkySdk

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        // Step 1: Create SDK instance (requires Context)
        sdk = NeuroSkySdk(this)

        // Step 2: Observe connection state
        lifecycleScope.launch {
            sdk.connectionState.collect { state ->
                when (state) {
                    ConnectionState.SCANNING -> updateStatus("Scanning...")
                    ConnectionState.CONNECTING -> updateStatus("Connecting...")
                    ConnectionState.CONNECTED -> updateStatus("Connected")
                    ConnectionState.ERROR -> updateStatus("Connection failed")
                    ConnectionState.DISCONNECTED -> updateStatus("Disconnected")
                }
            }
        }

        // Step 3: Connect and stream data
        lifecycleScope.launch {
            sdk.connect("MindWave Mobile")
            sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ) // Korea/USA
            // sdk.sendCommand(NeuroSkyCommand.NOTCH_50HZ) // Europe/China

            sdk.dataFlow.collect { data ->
                if (data.signalQuality == SignalQuality.NO_SIGNAL) {
                    updateStatus("No signal – adjust headset")
                    return@collect
                }
                attentionBar.progress = data.attention
                meditationBar.progress = data.meditation
                signalText.text = data.signalQuality.name
            }
        }
    }

    override fun onDestroy() {
        super.onDestroy()
        lifecycleScope.launch { sdk.disconnect() }
    }
}
```

Connecting by MAC address vs device name

`connect()` accepts either a device name or a Bluetooth MAC address:

```
// By device name (BLE scan required – slower, ~5 sec)
sdk.connect("MindWave Mobile")

// By MAC address (direct connect – faster, recommended for production)
sdk.connect("AA:BB:CC:DD:EE:FF")
```

Using the MAC address is faster and more reliable in production. Use device name during prototyping when you don't yet know the MAC.

7. EEG Data Model

`dataFlow` emits a `BrainWaveData` object for each packet received from the MindWave Mobile headset.

```
data class BrainWaveData(
    val timestamp: Long,           // System.currentTimeMillis() at receipt
    val poorSignal: Int,           // 0 = perfect, 200 = no contact
    val attention: Int,            // 0~100, eSense™ attention
    val meditation: Int,           // 0~100, eSense™ meditation
    val delta: Int,                // 0.5~2.75 Hz band power
    val theta: Int,                // 3.5~6.75 Hz band power
    val lowAlpha: Int,             // 7.5~9.25 Hz band power
    val highAlpha: Int,            // 10~11.75 Hz band power
    val lowBeta: Int,              // 13~16.75 Hz band power
    val highBeta: Int,             // 18~29.75 Hz band power
    val lowGamma: Int,             // 31~39.75 Hz band power
    val midGamma: Int,             // 41~49.75 Hz band power
    val rawEeg: List<Int>,         // 10 samples/packet, signed 16-bit, 512Hz
    val eyeBlink: Int,            // 0 = no blink, 1~255 = blink intensity
) {
    val signalQuality: SignalQuality // derived from poorSignal
}
```

Data update rates

Field(s)	Update rate	Notes
poorSignal	~1 Hz	Every packet
attention, meditation	~1 Hz	eSense™ computed once per second
delta through midGamma	~1 Hz	FFT computed once per second
rawEeg	512 Hz	10 samples per packet, ~51 packets/sec
eyeBlink	Event-driven	Only non-zero when blink detected

When `rawEeg` packets arrive, `attention`, `meditation`, and frequency band fields are `0` in that object — they only appear in the `eSense` packet which arrives separately. Filter by checking which fields are non-zero.

8. EEG Frequency Bands Explained

The ThinkGear chip performs a Fast Fourier Transform (FFT) on the raw EEG and outputs power in 8 frequency bands.

Important: values are relative, not absolute

The frequency band values are **relative power** in arbitrary units — not calibrated to physical units ($\mu\text{V}^2/\text{Hz}$). This means:

- You **cannot** compare values across different sessions or individuals in absolute terms
- You **can** compare values within a single session — e.g., "Delta rose 40% after eyes closed"
- Ratios** between bands are more meaningful than raw values

Band reference table

Field	Band	Frequency	Typical mental states
<code>delta</code>	δ Delta	0.5~2.75 Hz	Deep sleep, healing. High delta while awake = fatigue or poor signal.
<code>theta</code>	θ Theta	3.5~6.75 Hz	Drowsiness, daydreaming, creativity, REM sleep, deep meditation.
<code>lowAlpha</code>	α Low	7.5~9.25 Hz	Relaxed, unfocused, calm. Increases with eyes closed.
<code>highAlpha</code>	α High	10~11.75 Hz	Eyes-closed rest. Suppressed by active visual attention.
<code>lowBeta</code>	β Low	13~16.75 Hz	Active focus, alert thinking. The "work" band.
<code>highBeta</code>	β High	18~29.75 Hz	High arousal, stress, anxiety, intense cognition.
<code>lowGamma</code>	γ Low	31~39.75 Hz	Higher cognition, cross-modal perception, binding.
<code>midGamma</code>	γ Mid	41~49.75 Hz	Intense concentration. Elevated in expert meditators.

eSense™ Attention and Meditation

These are NeuroSky's **proprietary processed values** computed by the ThinkGear chip itself. The SDK receives them pre-computed.

Attention (0~100) — mental focus or concentration:

Range	Meaning
0	Not yet computed (startup, or no signal)

1~40	Low — distracted, relaxed, wandering mind
40~60	Neutral baseline
60~80	Moderate focus — engaged in task
80~100	High focus — strong active concentration

Meditation (0~100) — mental calmness or relaxation:

Range	Meaning
0	Not yet computed (startup, or no signal)
1~40	Low — active thinking, stress
40~60	Neutral baseline
60~80	Moderate relaxation
80~100	Deep calm — strong meditative state

eSense values require 10~20 seconds to stabilize after the headset is put on. Values of 0 at startup are normal.

Raw EEG

When Raw EEG is enabled, `rawEeg` contains 10 signed 16-bit samples per packet at 512 Hz:

```
// Enable raw EEG after connecting
sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)

sdk.dataFlow.collect { data ->
    data.rawEeg.forEach { sample ->
        // Each sample: -32768 to +32767
        // 512 Hz, 10 samples per packet
        plotSample(sample)
    }
}
```

Raw EEG is useful for custom FFT analysis, artifact detection, or research. It is **disabled by default** to reduce Bluetooth bandwidth.

9. Signal Quality

Signal quality is the most critical factor for usable data. Always check it before using attention, meditation, or band values.

PoorSignal values

Value	signalQuality	Reliability	Action
0	GOOD	Excellent	Use all data freely

1~50	FAIR	Acceptable	Minor noise, eSense still valid
51~199	POOR	Unreliable	Prompt user to adjust headset
200	NO_SIGNAL	No data	Headset not worn

Recommended check pattern

```

sdk.dataFlow.collect { data ->
    when (data.signalQuality) {
        SignalQuality.NO_SIGNAL -> {
            showMessage("Please put on the MindWave Mobile headset.")
            return@collect
        }
        SignalQuality.POOR -> {
            showMessage("Weak signal (${data.poorSignal}). Adjust the headset.")
            // Optionally still log, but don't drive UI decisions
            return@collect
        }
        SignalQuality.FAIR,
        SignalQuality.GOOD -> {
            // Data is usable – update UI
            updateAttentionUi(data.attention)
            updateMeditationUi(data.meditation)
        }
    }
}

```

Tips for improving signal quality

1. **Moisten the sensor** — a small drop of water on the forehead sensor significantly improves conductance
2. **Clean the forehead** — remove sunscreen, makeup, or sweat residue
3. **Adjust headset position** — center the sensor on FP1 (above the left eyebrow)
4. **Check the ear clip** — must make firm contact with the earlobe
5. **Wait 20~30 seconds** — after putting on the headset for signal to stabilize
6. **Avoid strong muscle movement** — clenching the jaw creates EMG artifact that looks like a signal-quality drop

10. Commands

After connecting, send control commands to configure the MindWave Mobile headset's behavior.

Notch filter

EEG signals are extremely low amplitude and easily contaminated by AC power-line noise. Send the notch filter command immediately after connecting:

Region	Grid frequency	Command
Korea	60 Hz	NeuroSkyCommand.NOTCH_60HZ

USA / Canada	60 Hz	NeuroSkyCommand.NOTCH_60HZ
Europe	50 Hz	NeuroSkyCommand.NOTCH_50HZ
China	50 Hz	NeuroSkyCommand.NOTCH_50HZ
Australia / UK	50 Hz	NeuroSkyCommand.NOTCH_50HZ

```

sdk.connect("MindWave Mobile")
sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ) // Korea/USA
// sdk.sendCommand(NeuroSkyCommand.NOTCH_50HZ) // Europe/China

```

Without the notch filter, you will likely see a large 50Hz or 60Hz artifact in raw EEG and elevated Beta band values.

Raw EEG streaming

```

// Enable raw EEG (disabled by default)
sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)

// Disable when no longer needed
sdk.sendCommand(NeuroSkyCommand.STOP_RAW_EEG)

```

eSense control

```

// Disable eSense if only raw EEG is needed
sdk.sendCommand(NeuroSkyCommand.STOP_ESENSE)

// Re-enable
sdk.sendCommand(NeuroSkyCommand.START_ESENSE)

```

All commands

Constant	Byte	Description
NeuroSkyCommand.NOTCH_60HZ	0x1C	Notch filter at 60 Hz
NeuroSkyCommand.NOTCH_50HZ	0x1B	Notch filter at 50 Hz
NeuroSkyCommand.START_RAW_EEG	0x15	Enable raw EEG stream
NeuroSkyCommand.STOP_RAW_EEG	0x16	Disable raw EEG stream
NeuroSkyCommand.START_ESENSE	0x17	Enable eSense output
NeuroSkyCommand.STOP_ESENSE	0x18	Disable eSense output

11. Simulator · Development Aid

SimulatorTransport is a supplementary development tool that generates synthetic EEG data for UI and logic testing. It implements Transport, so application code remains unchanged between Simulator and real-device sessions. For BLE and BT Classic validation, a physical MindWave Mobile 2 is always required.

When to use the Simulator (supplementary only)

- Early UI development · build and iterate on UI and data flows before hardware is available
- Predictable test data · use FOCUSED / RELAXED modes to validate attention and meditation UI
- Error-handling tests · POOR_SIGNAL mode verifies your reconnect and fallback logic
- CI/CD pipelines · run automated Espresso tests on build servers (not a substitute for device tests)

Note: The Simulator supports UI and logic testing only.

BLE and BT Classic connectivity must be tested with a physical MindWave Mobile 2 device.

```
import com.neurosky.sdk.simulator.SimulatorTransport // package: simulator, NOT transport

val simulator = SimulatorTransport()
simulator.setMode(SimulatorTransport.Mode.FOCUSED)

lifecycleScope.launch {
    simulator.connect("simulator") // any string accepted; CONNECTED after ~500 ms

    simulator.dataFlow.collect { data ->
        Log.d("SIM", "Attention: ${data.attention}, Meditation: ${data.meditation}")
    }
}
```

*stateFlow vs connectionState: SimulatorTransport (and all Transport implementations) expose stateFlow: Flow<ConnectionState> from the Transport interface. connectionState: StateFlow<ConnectionState> is a property of NeuroSkySdk only — it does **not** exist on SimulatorTransport or the Transport interface directly.*

```
// Wrong – connectionState does not exist on SimulatorTransport
simulator.connectionState.collect { } // compile error

// Correct – use stateFlow (Transport interface)
simulator.stateFlow.collect { state -> /* CONNECTED, DISCONNECTED, ... */ }
```

Simulator modes

Mode	Attention	Meditation	PoorSignal	Use case
'RANDOM'	0~100 (random)	0~100 (random)	0	General integration testing
'FOCUSED'	70~100	40~60	0	High-attention UI testing
'RELAXED'	20~50	70~100	0	High-meditation UI testing
'POOR_SIGNAL'	0	0	150~200	Signal loss and error handling

Switching modes at runtime

```
```kotlin
val simulator = SimulatorTransport()
```

```

simulator.setMode(SimulatorTransport.Mode.POOR_SIGNAL)

lifecycleScope.launch {
 simulator.connect("simulator")

 // Simulate signal recovery after 5 seconds
 delay(5000)
 simulator.setMode(SimulatorTransport.Mode.FOCUSED)

 simulator.dataFlow.collect { data -> /* ... */ }
}

```

## Using in ViewModel with dependency injection

```

// Swap between simulator and real SDK via DI or build flavor
class EegViewModel(
 private val transport: Transport // injected: NeuroSkySdk or SimulatorTransport
) : ViewModel() {

 val data: Flow<BrainWaveData> = transport.dataFlow
 val state: Flow<ConnectionState> = transport.stateFlow

 fun connect(address: String) {
 viewModelScope.launch {
 transport.connect(address)
 }
 }
}

```

## 12. Error Handling & Reconnection

### Connection errors

`connect()` is a suspend function — wrap in `try/catch` for production use:

```

lifecycleScope.launch {
 try {
 sdk.connect("MindWave Mobile")
 } catch (e: SecurityException) {
 // Bluetooth permission not granted
 showError("Bluetooth permission required.")
 } catch (e: Exception) {
 // BT adapter off, device not found, etc.
 showError("Connection failed: ${e.message}")
 }
}

```

### Auto-reconnect loop

`dataFlow` completes (collection ends) when the connection drops. Wrap in a retry loop for production:

```

lifecycleScope.launch {
 while (isActive) { // isActive = CoroutineScope is still alive
 try {
 sdk.connect("MindWave Mobile")
 sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)

 sdk.dataFlow.collect { data ->
 processData(data)
 }

 // dataFlow completed – headset turned off
 updateStatus("Disconnected. Reconnecting...")

 } catch (e: CancellationException) {
 break // ViewModel/lifecycle scope cancelled – exit cleanly
 } catch (e: Exception) {
 updateStatus("Error: ${e.message}. Retrying in 3s...")
 }

 delay(3000) // wait before retry
 }
}

```

### Handling NO\_SIGNAL without disconnecting

The Bluetooth connection can remain active while the electrode is not touching the skin. In this case, `signalQuality` becomes `NO_SIGNAL` but `dataFlow` keeps emitting. Handle this in your collector:

```

sdk.dataFlow.collect { data ->
 if (data.signalQuality == SignalQuality.NO_SIGNAL) {
 updateUi(connected = true, signalOk = false)
 return@collect
 }
 updateUi(connected = true, signalOk = true)
 processData(data)
}

```

## 13. Advanced Patterns

### ViewModel + StateFlow for Compose or XML UI

```

class EegViewModel(application: Application) : AndroidViewModel(application) {

 private val sdk = NeuroSkySdk(application)

 private val _attention = MutableStateFlow(0)
 private val _meditation = MutableStateFlow(0)
 private val _signal = MutableStateFlow(SignalQuality.NO_SIGNAL)
}

```

```

val attention: StateFlow<Int> = _attention.asStateFlow()
val meditation: StateFlow<Int> = _meditation.asStateFlow()
val signal: StateFlow<SignalQuality> = _signal.asStateFlow()
val state: StateFlow<ConnectionState> = sdk.connectionState

init {
 viewModelScope.launch {
 sdk.connect("MindWave Mobile")
 sdk.sendCommand(NeuroSkyCommand.NOTCH_60HZ)

 sdk.dataFlow.collect { data ->
 _signal.value = data.signalQuality
 _attention.value = data.attention
 _meditation.value = data.meditation
 }
 }
}

override fun onCleared() {
 viewModelScope.launch { sdk.disconnect() }
}
}

```

## Jetpack Compose integration

```

@Composable
fun EggScreen(viewModel: EggViewModel = viewModel()) {
 val attention by viewModel.attention.collectAsState()
 val meditation by viewModel.meditation.collectAsState()
 val signal by viewModel.signal.collectAsState()

 Column(modifier = Modifier.padding(16.dp)) {
 Text("Signal: $signal")
 LinearProgressIndicator(progress = attention / 100f)
 Text("Attention: $attention")
 LinearProgressIndicator(progress = meditation / 100f)
 Text("Meditation: $meditation")
 }
}

```

## Buffering 1 second of raw EEG for custom analysis

```

// Raw EEG arrives as 10 samples per packet.
// Accumulate 512 samples for 1 second of data at 512 Hz.

val buffer = mutableListOf<Int>()

sdk.sendCommand(NeuroSkyCommand.START_RAW_EEG)

```



```

sdk.dataFlow.collect { data ->
 if (data.rawEeg.isEmpty()) return@collect

 buffer.addAll(data.rawEeg)

 if (buffer.size >= 512) {
 val oneSecond = buffer.take(512).toIntArray()
 buffer.subList(0, 512).clear()

 // Run your own FFT or signal processing
 val spectrum = MyFft.compute(oneSecond, sampleRate = 512)
 displaySpectrum(spectrum)
 }
}

```

## Recording session data to a file

```

val filename = "eeg_${System.currentTimeMillis()}.csv"
val file = File(getExternalFilesDir(null), filename)

file.bufferedWriter().use { writer ->
 writer.write("timestamp,attention,meditation,poor_signal,delta,theta," +
 "low_alpha,high_alpha,low_beta,high_beta,low_gamma,mid_gamma\r\n")

 sdk.dataFlow
 .filter { it.attention > 0 || it.meditation > 0 } // skip raw-only packets
 .collect { d ->
 writer.write("${d.timestamp},${d.attention},${d.meditation},${d.poorSignal}," +
 "${d.delta},${d.theta},${d.lowAlpha},${d.highAlpha}," +
 "${d.lowBeta},${d.highBeta},${d.lowGamma},${d.midGamma}\r\n")
 }
}

```

## Background service with foreground notification

For applications that need to stream EEG data while the app is in the background:

```

class EggForegroundService : Service() {

 private val sdk by lazy { NeuroSkySdk(this) }
 private val scope = CoroutineScope(SupervisorJob() + Dispatchers.IO)

 override fun onStartCommand(intent: Intent?, flags: Int, startId: Int): Int {
 startForeground(NOTIFICATION_ID, buildNotification())

 scope.launch {
 sdk.connect("MindWave Mobile")
 sdk.dataFlow.collect { data ->
 broadcastData(data) // send to Activity via LocalBroadcastManager
 }
 }
 }
}

```

```

 }

 return START_STICKY
}

override fun onDestroy() {
 scope.cancel()
 super.onDestroy()
}
}

```

## 14. Troubleshooting

### Connection issues

Symptom	Likely cause	Solution
SecurityException on connect()	Bluetooth permission not granted at runtime	Request BLUETOOTH_SCAN + BLUETOOTH_CONNECT (API 31+) or ACCESS_FINE_LOCATION (API 23–30)
connect() hangs indefinitely	BLE scan timeout	Use findDeviceAddress() first to resolve the MAC address, then pass the MAC to connect() for faster connection.
BT Classic connect fails	Device not paired	Open Android Settings → Bluetooth → pair "MindWave Mobile" manually
Connection drops after a few minutes	Android BLE background scan kill	Use a Foreground Service to prevent the OS from killing the connection
dataFlow stops emitting after screen off	Background execution limit	Same — use a Foreground Service

### Signal quality issues

Symptom	Likely cause	Solution
NO_SIGNAL immediately after connecting	Electrode not touching forehead	Adjust headset and press sensor firmly to skin
POOR signal persists after 30 seconds	Dry skin or dirty sensor	Wet the sensor tip with water; clean forehead
attention and meditation always 0	eSense needs warm-up	Wait 20–30 seconds after achieving GOOD signal
50Hz or 60Hz spike in raw EEG	No notch filter	Call sendCommand(NeuroSkyCommand.NOTCH_60HZ) after connecting
rawEeg always empty	START_RAW_EEG not called	Call sendCommand(NeuroSkyCommand.START_RAW_EEG) after connecting

## Build issues

Symptom	Likely cause	Solution
Could not resolve com.github.nsk- bci:mindwave-sdk-android	JitPack not in repository list	Add maven { url = uri("https://jitpack.io") } to settings.gradle.kts
Unresolved reference: NeuroSkySdk	Missing import	Add import com.neurosky.sdk.NeuroSkySdk
Unresolved reference: SimulatorTransport	Wrong import	Use import com.neurosky.sdk.simulator.SimulatorTransport (package is simulator, not transport)
Unresolved reference: TransportMode	Old doc typo	The correct enum is TransportType. Use TransportType.BLE / TransportType.BT_CLASSIC
Unresolved reference: connectionState on simulator	Wrong API	connectionState belongs to NeuroSkySdk. Use simulator.stateFlow instead
Unresolved reference: alphaLow / betaHigh	Wrong field name order	Field names are lowAlpha, highAlpha, lowBeta, highBeta, lowGamma, midGamma — modifier first
First build very slow	JitPack building from source	Normal — only happens once; subsequent builds use cache

## Data issues

Symptom	Likely cause	Solution
dataFlow emits nothing even though connected	dataFlow captured before connect()	Move sdk.dataFlow.collect to inside the same coroutine after connect() returns
BLE connected but data always 0, NO_SIGNAL forever	Notch filter not set, or eSense not started	Call sendCommand(NeuroSkyCommand.NOTCH_60HZ) after connecting
Connection takes 5+ seconds	BLE scan by device name	Use findDeviceAddress() first, cache the MAC, then pass MAC to connect()

## 15. Testing

The SDK ships with a unit test suite for `ThinkGearParser` — the packet parser that runs identically on both BLE and BT Classic transports. These tests require no hardware or Bluetooth adapter and run on the JVM directly.

### Running the tests

```
./gradlew :sdk:test
```

Test results are written to:

```
sdk/build/reports/tests/testDebugUnitTest/index.html
```

## What is covered

Test	Description
parseEsense_0xEA	Attention, Meditation, PoorSignal extraction
parseEsense_0xEA_tooShort	Short packet → returns null
parseEsense_0xEB	Delta, Theta, LowAlpha, HighAlpha extraction
parseEsense_0xEC	LowBeta, HighBeta, LowGamma, MidGamma extraction
parseRawEeg_returns10Samples	20-byte raw EEG → 10 signed int samples
parseRawEeg_signedConversion	Values > 32768 converted to negative
parseRawEeg_tooShort	Short packet → returns null
parse_unknownUuid	Unknown UUID → returns null
parseByte_validPacket	BT Classic serial packet — Attention/Meditation
parseByte_invalidChecksum	Wrong checksum → returns null
parseByte_poorSignalCode	BT Classic PoorSignal (code 0x02)
signalQuality_*	200/100/25/0 → NO_SIGNAL/POOR/FAIR/GOOD

## Test location

```
sdk/src/test/kotlin/com/neurosky/sdk/parser /
└── ThinkGearParserTest.kt
```

## 16. API Reference

### NeuroSkySdk

Main entry point. Manages BLE/BT Classic transport selection and lifecycle.

```
class NeuroSkySdk(context: Context)
```

Member	Type	Description
connectionState	StateFlow<ConnectionState>	Current connection state; hot Flow, always has a value
dataFlow	Flow<BrainWaveData>	Returns <code>activeTransport.dataFlow</code> at call time — collect <b>after</b> <code>connect()</code>
connect(deviceAddress, transport)	suspend fun	Connects via the given <code>TransportType</code> . Default: <code>TransportType.BLE</code> . No automatic

		fallback
<code>disconnect()</code>	suspend fun	Gracefully closes the active transport
<code>sendCommand(cmd: Byte)</code>	suspend fun	Sends a control byte to the headset
<code>findDeviceAddress(deviceName, timeoutMs)</code>	suspend fun	BLE scan returning MAC address, or null on timeout

**dataFlow timing:** `sdk.dataFlow` is a property getter — each access returns the current `activeTransport.dataFlow`. Capturing `sdk.dataFlow` into a variable before `connect()` will bind the flow to the pre-connect transport state. Always collect inside the same coroutine that calls `connect()`, or after `connectionState` emits `CONNECTED`.

**TransportType not TransportMode:** The enum is `TransportType` with values `BLE` and `BT_CLASSIC`. `TransportMode` does not exist in the API.

## BrainWaveData

Immutable data class emitted by `dataFlow`.

Field	Type	Range	Description
<code>timestamp</code>	Long	—	<code>System.currentTimeMillis()</code> at receipt
<code>poorSignal</code>	Int	0~200	0 = perfect contact, 200 = no contact
<code>attention</code>	Int	0~100	eSense™ attention (0 = not computed)
<code>meditation</code>	Int	0~100	eSense™ meditation (0 = not computed)
<code>delta</code>	Int	0~∞	Delta band power, 0.5~2.75 Hz
<code>theta</code>	Int	0~∞	Theta band power, 3.5~6.75 Hz
<code>lowAlpha</code>	Int	0~∞	Low Alpha, 7.5~9.25 Hz
<code>highAlpha</code>	Int	0~∞	High Alpha, 10~11.75 Hz
<code>lowBeta</code>	Int	0~∞	Low Beta, 13~16.75 Hz
<code>highBeta</code>	Int	0~∞	High Beta, 18~29.75 Hz
<code>lowGamma</code>	Int	0~∞	Low Gamma, 31~39.75 Hz
<code>midGamma</code>	Int	0~∞	Mid Gamma, 41~49.75 Hz
<code>rawEeg</code>	List<Int>	-32768~32767	512Hz ADC samples (10 per packet)
<code>eyeBlink</code>	Int	0~255	Eye blink intensity; 0 = no blink
<code>signalQuality</code>	SignalQuality	enum	Derived from <code>poorSignal</code>

## ConnectionState

```
enum class ConnectionState { DISCONNECTED, SCANNING, CONNECTING, CONNECTED, ERROR }
```

Value	Meaning
DISCONNECTED	No active connection
SCANNING	Scanning for the target device
CONNECTING	Device found, establishing connection
CONNECTED	Data stream active
ERROR	Connection attempt failed

## SignalQuality

Derived from `BrainWaveData.poorSignal` .

Value	poorSignal	Reliability	Recommended action
GOOD	0	Excellent	Use all data
FAIR	1~50	Acceptable	Use data; minor noise present
POOR	51~199	Unreliable	Prompt user to adjust headset
NO_SIGNAL	200	No data	Prompt user to put on headset

## Transport (interface)

Common interface implemented by `BleTransport` , `BtClassicTransport` , and `SimulatorTransport` .

```
interface Transport {
 val dataFlow: Flow<BrainWaveData>
 val stateFlow: Flow<ConnectionState>
 suspend fun connect(deviceAddress: String)
 suspend fun disconnect()
 suspend fun sendCommand(cmd: Byte)
}
```

## SimulatorTransport

```
class SimulatorTransport : Transport
```

Member	Description
<code>setMode(mode: Mode)</code>	Change simulation mode; takes effect on next emitted packet
<code>Mode.RANDOM</code>	Random Attention and Meditation values each tick

Mode.FOCUSED	Attention 70100, Meditation 4060
Mode.RELAXED	Attention 2050, Meditation 70100
Mode.POOR_SIGNAL	PoorSignal 150~200, Attention 0, Meditation 0

## NeuroSkyCommand

**object** NeuroSkyCommand

Constant	Byte	When to use
NOTCH_60HZ	0x1C	Power grid is 60 Hz (Korea, USA)
NOTCH_50HZ	0x1B	Power grid is 50 Hz (Europe, China)
START_RAW_EEG	0x15	Enable raw EEG waveform (disabled by default)
STOP_RAW_EEG	0x16	Disable raw EEG waveform
START_ESENSE	0x17	Enable Attention/Meditation (enabled by default)
STOP_ESENSE	0x18	Disable Attention/Meditation

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